

# Yotam Sechayk

Ph.D. Candidate · The University of Tokyo · Human-Computer Interaction & Accessibility

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## Research Interests

I am a human-computer interaction and accessibility researcher with an interdisciplinary background. My work focuses on **making visual communication visually accessible** for low-vision people. I study how low-vision people use their residual vision to access lecture videos, data charts, and the web. I follow user-centered and ability-based design, working with low-vision people throughout the design process and building on the abilities they have. As a low-vision person myself, I design interaction techniques and AI-powered systems that *enhance*, rather than replace, existing visual strategies. I have published **six full papers** at CHI, ASSETS, and UIST, and my work has received **three paper awards**, including an ASSETS Honorable Mention.

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## Education

- 2024.04 – Present    **The University of Tokyo** · Tokyo, Japan  
Ph.D. in Creative Informatics, Graduate School of Information Science and Technology (expected 2027.03)  
Advisor: Prof. Takeo Igarashi · Co-advisor: Prof. Ariel Shamir (Reichman University)  
Dissertation (in progress): *Visual Communication for Low-Vision Information Access*
- 2022.04 – 2024.03    **The University of Tokyo** · Tokyo, Japan  
M.Sc. in Creative Informatics, Graduate School of Information Science and Technology  
Advisor: Prof. Takeo Igarashi  
Thesis: *Smart Replay: Visual-Temporal Accessibility in e-learning Videos*
- 2017.04 – 2017.09    **Waseda University** · Tokyo, Japan  
Exchange Student, School of International Liberal Studies
- 2014.10 – 2018.09    **Tel Aviv University** · Tel Aviv, Israel  
B.Sc., Double Major in Computer Science and Film & Television  
Graduated with honors (magna cum laude; GPA 93/100).

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## Research Experience

- 2022.04 – Present **Igarashi Lab** · The University of Tokyo  
Advisor: Prof. Takeo Igarashi · Co-advisor: Prof. Ariel Shamir (Reichman University)
- Lead a SoftBank Beyond AI CIP project on web accessibility with AI agents for low-vision users (2026 – present). Manage the research budget, supervise a research intern, and coordinate development with the SoftBank team.
  - Built *Improving Low-Vision Chart Accessibility via On-Cursor Visual Context*, comparing two pointer-based context methods (Dynamic Context, Mini-map) with  $N=22$  LVI. [C.5]
  - Designed *VeasyGuide*, a tool that detects instructor actions in lecture videos and dynamically highlights and magnifies them for low-vision learners. Evaluated with 8 LV and 8 sighted participants. Open source and available online at [veasyguide.github.io/app](https://veasyguide.github.io/app). [C.3]
  - Earlier and smaller projects on visual-temporal accessibility of e-learning videos, chart reading, and creative tools: *Graph Guide* [D.4], *ShowMe* [D.3], *SmartLearn* [W.1], *Smart Replay* [D.1], and *MyStoryKnight* [D.2].
- 2024.08 – 2024.10 **University of Texas at Austin** · Austin, TX  
Visiting Researcher  
Advisor: Prof. Amy Pavel  
Recruited participants and ran the user studies for *VeasyGuide*. [C.3] Continued the collaboration after the visit on projects including *Task Mode*. [C.2]
- 2023.10 – 2024.03 **Honda Research Institute Japan** · Saitama, Japan  
Research Intern, Human-Robot Interaction  
Supervisor: Dr. Christian Arzate Cruz  
Augmented facial-expression data with 3D morphable models to improve arousal–valence prediction in HRI settings. [C.1]

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## Work Experience

- 2018.03 – 2020.10 **Cadence Design Systems** · Israel  
Software Engineer, Multicore Engine Team  
Developed semiconductor verification tools utilizing parallel computing.
- 2015.03 – 2017.03 **Tel Aviv University** · Tel Aviv, Israel  
Head of Technology, Disability Support  
Investigated software and hardware solutions to assist students with disabilities and supported their adoption in coursework.

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## Publications

Seven full papers at international venues (CHI, ASSETS, UIST, RO-MAN), a UIST Doctoral Symposium paper, eight posters, demos, and workshop papers, and four papers at Japanese venues (WISS, INTERACTION). Three papers received awards, including an ASSETS Honorable Mention and a WISS Best Paper Award.

\* indicates equal contribution. My name is underlined. Entries are numbered chronologically (1 = earliest).

### International Conference Papers

- C.7 LandmarkLens: Predicting and Presenting Effective Landmarks for Mixed-Reality Urban Exploration**  
Chu Li, Yotam Sechayk, Jared Hwang, Jon E. Froehlich, Takeo Igarashi  
UIST 2026: ACM Symposium on User Interface Software and Technology.
- C.6 Ups and Downs: Low-Vision Perspective on Smart Elevators and Industry Design Practices**  
Masaki Sawada, Yotam Sechayk, Manabi Miyagi  
ASSETS 2026: ACM SIGACCESS Conference on Computers and Accessibility (experience report).
- C.5 Improving Low-Vision Chart Accessibility via On-Cursor Visual Context**  
Yotam Sechayk, Hennes Rave, Max Rädler, Mark Colley, Zhongyi Zhou, Ariel Shamir, Takeo Igarashi  
CHI 2026: ACM Conference on Human Factors in Computing Systems.
- C.4 No Pixel Left Behind: Filling Gaps in Anime Colorization**  
Masahiro Kono, Akinobu Maejima, Yuki Koyama, Yotam Sechayk, Takeo Igarashi  
CHI 2026: ACM Conference on Human Factors in Computing Systems.
- C.3 VeasyGuide: Personalized Visual Guidance for Low-vision Learners on Instructor Actions in Presentation Videos**  
Yotam Sechayk, Ariel Shamir, Amy Pavel, Takeo Igarashi  
ASSETS 2025: ACM SIGACCESS Conference on Computers and Accessibility.  
 **Honorable Mention Award**
- C.2 Task Mode: Dynamic Filtering for Task-Specific Web Navigation using LLMs**  
Ananya Gubbi Mohanbabu, Yotam Sechayk, Amy Pavel  
ASSETS 2025: ACM SIGACCESS Conference on Computers and Accessibility.
- C.1 Data Augmentation for 3DMM-based Arousal-Valence Prediction for HRI**  
Christian Arzate Cruz\*, Yotam Sechayk\*, Takeo Igarashi, Randy Gomez  
RO-MAN 2024: IEEE International Conference on Robot and Human Interactive Communication.

### Doctoral Symposium

- DS.1 Visual Communication for Low-Vision Information Access: From Personalized Highlights to AI Agents**  
Yotam Sechayk  
UIST Adjunct 2026: Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology.

### Workshops, Posters, Demos, and Extended Abstracts

- W.8 Cyber-Physical Systems for Accessibility and Ability Augmentation: Bridging Diverse Communities**  
 Shuchang Xu\*, Riku Arakawa\*, Mina Huh\*, Nandi Zhang\*, Tianyu Zhang\*, Wazeer Zulfikar\*, Ruei-Che Chang\*, Yotam Sechayk\*, Huamin Qu, Amy Pavel, Franklin Mingzhe Li, Yukang Yan, Brian A. Smith, Pattie Maes  
 UIST Adjunct 2026: Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (workshop proposal).
- W.7 PreView: Rehearsing Real Routes in VR to Support Independent Navigation with Low-Vision**  
Yotam Sechayk, Xinyun Cao, Dhruv Jain, Takeo Igarashi  
 UIST Adjunct 2026: Adjunct Proceedings of the ACM Symposium on User Interface Software and Technology (demo).
- W.6 See What I Mean: A Technology Probe of AI Assistants for Chart Exploration with Low Vision**  
Yotam Sechayk, Lucy Jiang, Yuhang Zhao, Takeo Igarashi  
 ASSETS 2026: ACM SIGACCESS Conference on Computers and Accessibility (poster).
- W.5 Exploring the Role of User Comments Throughout the Stages of Video-Based Task-Learning**  
 Nayoung Kim\*, Yotam Sechayk\*, Zhongyi Zhou, Takeo Igarashi  
 CHI EA 2026: Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (poster).
- W.4 A Longitudinal Autoethnography of Email Access for a Professional with Chronic Illness and ADHD: Preliminary Insights**  
 Veronica Pimenova, Yotam Sechayk, Fabricio Murai, Andrew Hundt, Shiri Dori-Hacohen  
 ASSETS 2025: ACM SIGACCESS Conference on Computers and Accessibility (poster).
- W.3 Confidence Estimation of Few-shot Patch-based Learning for Anime-style Colorization**  
 Yuexiang Ji, Akinobu Maejima, Yotam Sechayk, Yuki Koyama, Takeo Igarashi  
 SIGGRAPH 2025 Posters.
- W.2 ImprovMate: Multimodal AI Assistant for Improv Actor Training**  
 Riccardo Drago, Yotam Sechayk, Mustafa Doga Dogan, Andrea Sanna, Takeo Igarashi  
 DIS 2025 Companion: ACM Conference on Designing Interactive Systems (work-in-progress).
- W.1 SmartLearn: Visual-Temporal Accessibility for Slide-based e-learning Videos**  
Yotam Sechayk, Ariel Shamir, Takeo Igarashi  
 CHI EA 2024: Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems (late-breaking work).

### **Domestic Conference Papers (Japan)**

- D.4 Graph Guide: 低視力者支援のためのセマンティック Focus+Context グラフ表示**  
 Graph Guide: Semantic Focus+Context Graph Display for Low-Vision Users  
Yotam Sechayk, Yuan Li, Hennes Rave, Mark Colley, Ariel Shamir, Takeo Igarashi  
 WISS 2025: 第 33 回インタラクティブシステムとソフトウェアに関するワークショップ, IPSJ (in Japanese).  
 **Best Paper Award (最優秀論文賞)**

- D.3 ShowMe: 対話的な強調表示と拡大表示によるプレゼンテーションビデオの視覚的アクセシビリティの改善**  
 ShowMe: Interactive Highlighting and Magnification for Improving Visual Accessibility of Presentation Videos  
Yotam Sechayk, Ariel Shamir, Takeo Igarashi  
 WISS 2024: 第 32 回インタラクティブシステムとソフトウェアに関するワークショップ, IPSJ (in Japanese).
- D.2 MyStoryKnight: A Character-drawing Driven Storytelling System Using LLM Hallucinations**  
Yotam Sechayk, Gabriela A. Penarska, Ingrid A. Randsalu, Christian Arzate Cruz, Takeo Igarashi  
 INTERACTION 2024, IPSJ (poster/demo).
- D.1 Smart Replay: e ラーニング動画における視覚的・時間的アクセシビリティの向上**  
 Smart Replay: Improving Visual and Temporal Accessibility in E-learning Videos  
Yotam Sechayk, Ariel Shamir, Takeo Igarashi  
 WISS 2023: 第 31 回インタラクティブシステムとソフトウェアに関するワークショップ, IPSJ (in Japanese).  
 **Helpfeel Award for Corporate Excellence**

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## Patents

Published **Data Collection Method and Data Collection Device**  
 Christian Arzate Cruz, Randy Gomez, Yotam Sechayk  
 WO 2026/053487 A1 (PCT/JP2025/015777), published March 2026. Assignee: Honda Motor Co., Ltd.  
 Invented during internship at Honda Research Institute Japan. [C.1]

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## Awards & Honors

- 2025 **Best Paper Award** (*Saiyushu Ronbunsho*) · WISS 2025  
 Awarded for Graph Guide. [D.4]
- 2025 **Honorable Mention Award** · ACM ASSETS 2025  
 Awarded for VeasyGuide. [C.3]
- 2023 **Helpfeel Award for Corporate Excellence** · WISS 2023  
 Awarded for Smart Replay. [D.1]
- 2018 **Graduated with Honors** (magna cum laude) · Tel Aviv University

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## Grants & Fellowships

Co-wrote three funded research proposals that secured about USD 225,000 (JPY 35,500,000) for the lab. Received two MEXT scholarships worth about USD 80,000 (JPY 12,800,000) in tuition and stipends.

- 2024 – Present **MEXT Scholarship for Ph.D. Studies in Japan** · Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT)  
 Full tuition and a monthly stipend of JPY 148,000 for the doctoral program (about USD 49,000 in total).
- 2026 – 2027 **Softbank Beyond AI CIP Grant** · Softbank  
 JPY 20,000,000 (about USD 127,000) for web accessibility with AI agents (Jun 2026 – Mar 2027).  
 Co-wrote the proposal; project lead. PI: Prof. Takeo Igarashi.
- 2025 **Softbank Beyond AI Research Grant** · Softbank  
 JPY 10,000,000 (about USD 63,000). Co-wrote the proposal. PI: Prof. Takeo Igarashi.

- 2023 **Google Award for Inclusion Research** · Google  
USD 35,000 for “Enhance E-Learning Accessibility: Exploring AI Supported Temporal Approaches for Inclusive Online Education.” Co-wrote the proposal; PI: Prof. Takeo Igarashi.
- 2022 – 2024 **MEXT Scholarship for Master’s Studies in Japan** · Japanese Ministry of Education, Culture, Sports, Science and Technology (MEXT)  
Full tuition and a monthly stipend of JPY 146,000 for the master’s program (about USD 32,000 in total).

## Invited Talks & Research Visits

Invited talks at KAIST, Keio University, Hokkaido University, Miraikan, UT Austin, and Microsoft Israel, and research visits to University College London, the University of Washington, and UT Austin.

- 2026.04 **Research Visit** · University College London (UCL), London, UK  
Host: Dr. Mark Colley (UCL Interaction Centre).
- 2026.02 **Increasing Access in Online Education through Personalized Visual Highlights**  
Invited Talk, KIXLAB, KAIST, Daejeon, South Korea. Host: Prof. Juho Kim.
- 2026.01 **Increasing Access in Online Education through Personalized Visual Highlights**  
Invited Talk, Embodied Media Project, Keio University Graduate School of Media Design (KMD), Yokohama, Japan. Host: Dr. Takatoshi Yoshida.
- 2025.12 **Increasing Access in Online Education through Personalized Visual Highlights**  
Invited Talk, HCI Lab, Hokkaido University, Sapporo, Japan. Host: Prof. Daisuke Sakamoto.
- 2025.11 **Highlights and Context: Augmenting Sight for Low-vision People**  
Invited Talk, Accessibility Lab, Miraikan (National Museum of Emerging Science and Innovation), Tokyo, Japan. Host: Dr. Hironobu Takagi.
- 2025.09 – 2025.11 **Research Visit** · University of Washington, Seattle, WA  
Host: Prof. Jon E. Froehlich (Makeability Lab).
- 2024.10 **Awareness-(less) based Accessibility**  
Invited Talk, University of Texas at Austin. Host: Prof. Amy Pavel.
- 2024.08 – 2024.10 **Research Visit** · University of Texas at Austin, Austin, TX  
Host: Prof. Amy Pavel.
- 2023.05 **Accessibility and Usability in Online Learning Videos**  
Invited Talk, Microsoft Israel. Host: Deep Learning Team.

## Conference Presentations

- 2026 ACM CHI 2026 [C.5] · ACM ASSETS 2026 [C.6] (joint) · ACM UIST 2026 Doctoral Symposium [DS.1]
- 2025 ACM ASSETS 2025 [C.3] · WISS 2025 [D.4]
- 2024 WISS 2024 [D.3]
- 2023 WISS 2023 [D.1]

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## Teaching

- 2026 – Present **Creative Informatics Seminar (Master's and Doctoral)**  
Teaching Assistant · Graduate School of Information Science and Technology, The University of Tokyo (4860-2008/2009)  
Led a complete restructuring of the course format, backed by student feedback and prior research on interdisciplinary research and research communication. The new format launched in April 2026.
- Introduced weekly poster sessions to give students more opportunities to interact.
  - Created online mechanisms for students to give each other feedback.
  - Received positive faculty and student feedback and strong participation.
- 2026 **Accessibility in HCI and the AI Revolution**  
Guest Lecture · Undergraduate Human-Computer Interaction (0510071), The University of Tokyo  
Instructor of record: Prof. Takeo Igarashi.
- 2025 **Accessibility in HCI and the AI Revolution**  
Guest Lecture · Undergraduate Human-Computer Interaction (0510071), The University of Tokyo  
Instructor of record: Prof. Takeo Igarashi.

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## Mentoring & Advising

Mentored ten students and junior researchers, from undergraduates to Ph.D. students, at institutions in Japan, the US, Israel, Italy, and Korea. Five mentees co-authored papers from this work.

- 2026 – present **Faith Luo** · Research Intern on web accessibility with AI agents (Software Engineer, thatgamecompany)
- 2026 – present **Ritsu Ueda** · Georgia Institute of Technology
- 2026 – present **Masaki Sawada** · Tsukuba University of Technology. [C.6]
- 2026 **Koyo Tanaka** · University of California, San Diego
- 2025 – present **Eleyah Melamed** · Reichman University
- 2025 – present **Xinyun Cao** · University of Michigan. [W.7]
- 2024 – present **Veronica Lewis** · Ph.D., George Mason University
- 2024 – present **Veronica Pimenova** · University of Michigan. [W.4]
- 2024 – 2025 **Riccardo Drago** · M.Sc., Polytechnic University of Turin. [W.2]
- 2024 **Nayoung Kim** · Undergraduate, DGIST. [W.5]

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## Academic Service

### Organizing

- 2026 – 2027 **Accessibility Chair** · ACM CHI 2027 Organizing Committee

### Reviewing

- 2026 ACM CHI 2026 · ACM UIST 2026 · ACM MobileHCI 2026
- 2025 ACM CHI 2025 · ACM UIST 2025 · ACM DIS 2025 · ACM IDC 2025

2024 ACM CHI 2024 · ACM UIST 2024

Special Recognition for Outstanding Reviews: CHI 2025 (×2), UIST 2025, CHI 2026.

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## Outreach & Media

- 2026.07 **School Visit** · Tsukuba Special Needs Education School for the Visually Impaired  
Introduced our educational accessibility tools to teachers and discussed collaboration, with Dr. Mika Oki and Prof. Takeo Igarashi. July 28, 2026.
- 2026.05 **The University of Tokyo Visit for Low-Vision Students from Tsukuba University of Technology**  
Organized and hosted four students with low vision and Prof. Makoto Kobayashi, including an introduction to our research. May 7, 2026.
- 2025 *AT and Me: An Interview with Yotam Sechayk*  
Perkins School for the Blind · Paths to Technology · interview by Veronica Lewis, August 2025.

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## Languages

Hebrew (native) · English (fluent) · Japanese (advanced).